

HAT-P-54b

R-1734

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-54_180223 · created 2026-08-02T17:55:08+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458172.63195 +/- 0.00043 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.15 +/- 0.04
Transit depth (Rp/Rs)^2	2.26 +/- 1.19 [%]
Orbital Inclination (inc)	84.42 +/- 1.94
Airmass coefficient 1 (a1)	1.576 +/- 0.037
Airmass coefficient 2 (a2)	-0.45 +/- 0.25
Scatter in the residuals of the lightcurve fit is	2.31 %
Best Comparison Star	None
Optimal Aperture	2.81
Optimal Annulus	8.38
Transit Duration (day)	0.043 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.15 ± 0.04 vs 0.1572 (deviation 0.18σ)
Duration fit vs published	0.043 d vs 0.075 d (deviation 2.0σ)
Mid-time O–C	-0.8 min
Depth SNR	3.8
Residual scatter	2.31 %

Light curve

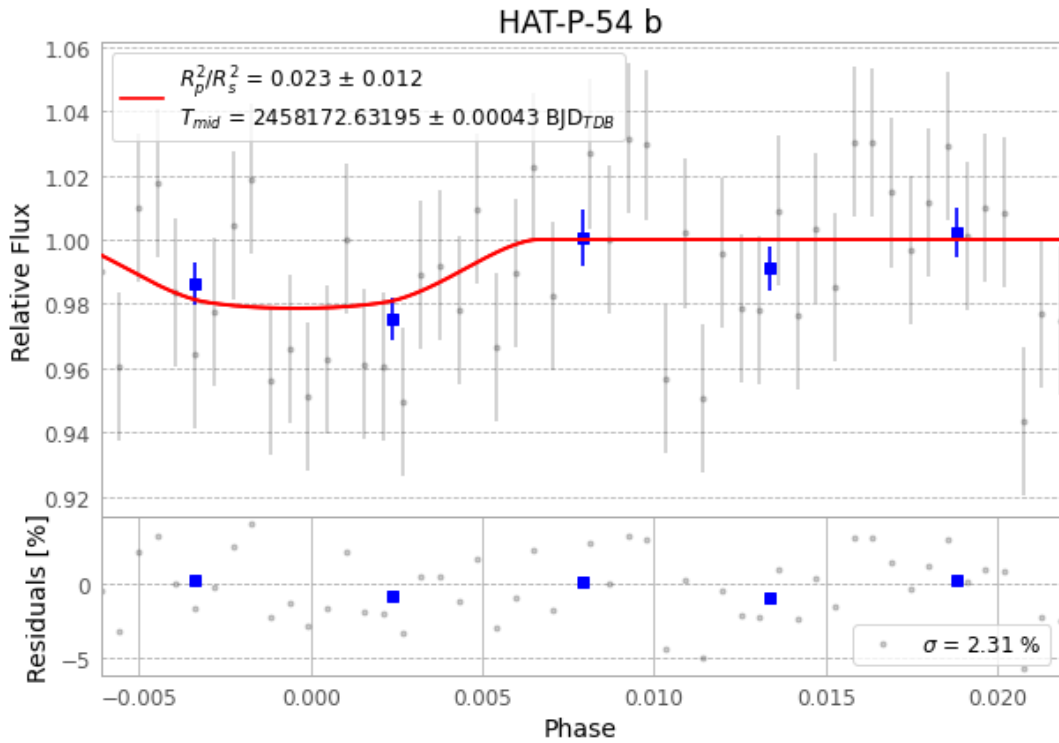


Figure 1 — FinalLightCurve_HAT-P-54 b_2018-02-22.png (SHA-256 70481c92273f5e52...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [409, 268], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 fbe8c9851f24e071...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

52 FITS frames (34 MB), HATP-54180223023012.FITS → HATP-54180223050312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-54-180223.log (e1cf4fc9039f...), FinalLightCurve_HAT-P-54 b_2018-02-22.png (70481c92273f...), FinalLightCurve_HAT-P-54 b_2018-02-22.csv (e01b0b5b4c62...)

Compute resources

Wall time 260.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 17:55Z.